PRODUCT DESCRIPTION — FOR DEVELOPER HANDOFF

Group Scheduler

A When2meet-style scheduler: the organizer creates one event, shares one link, everyone drag-paints when they are free, and the group heatmap shows the overlap at a glance.

> **Build decision — read this first:** The reference implementation is **When2meet**. Copy its interaction model, not a generic poll or form. If a feature in this document conflicts with how When2meet behaves, When2meet wins, unless the item is listed as an intentional improvement in Section 8.

Executive summary

This product solves scheduling an online meeting for a small group (about 5 to 20 people) spread across time zones. Chat polls cap the number of options and hide detail. Forms collect answers privately, so no one can see who is free when, and neither converts times reliably — a vote for “7:00 PM” can mean two different moments, and across the date line, two different days.

When2meet solves the core problem with one idea: a shared grid you paint, not a list you vote on. This product reproduces that idea faithfully — drag-paint availability in 15-minute cells, no accounts, no Save button, and a green heatmap where the darkest block is the answer — and adds only the correctness When2meet is often criticised for: reliable time zone conversion and painting that works properly on phones.

Primary use case

A recurring online small-group meeting — for example, a men’s small group scheduling its next meeting across a two-week window, with members in the United States and Indonesia.

Example event

Specific dates: 14 consecutive days, Nov 2–15. Time window: No earlier than 12:00 AM, No later than 11:00 PM — the full day. Every participant paints every 15-minute block when they are free.

Document type: Product description / mini product requirements document (PRD)

Audience: Software developer or development team building the product

Reference implementation: When2meet (when2meet.com) — its create flow, sign-in, painting, and heatmap

Status: Revised draft for handoff — open questions are listed in Section 9

How to use this document: Sections 4 and 9 are the build contract. Section 2.1 is the When2meet mapping to check every screen against.

1. The problem

A group organizer today typically tries a chat poll or a form. Both break down for a two-week, all-hours, cross-timezone poll:

- **Too many options for a chat poll.** WhatsApp polls cap the number of options, so 14 days — let alone 14 days combined with times — cannot fit in a single poll. Splitting the poll loses the combined picture.
- **Answers are hidden.** In a form, each respondent submits alone and cannot see what others selected. The group cannot self-organise around emerging overlap.
- **Ticking boxes is the wrong gesture.** Availability is a range, not a set of isolated choices. People need to mark “I am free from about 6 to 9” in one motion, including partial hours — which is why When2meet uses painting in 15-minute cells instead of checkboxes.
- **Time zones are not handled.** 7:00 PM Pacific is the next calendar morning in Western Indonesia Time (WIB), so both the hour and the day shift. A poll spanning a daylight-saving change must also stay correct on both sides of that change.

The result is predictable: answers that cannot be compared directly, manual follow-up in chat, and a real risk of announcing a time that is wrong for part of the group.

2. Who it is for

User	What they need
Organizer	Create an event in under a minute, share it with one link, watch the heatmap fill in, and read the best time off the darkest block.
Participant	Open a link with no account, type a name, paint free times in their own local time, and immediately see the group result beside their own grid.
Whole group	One shared, always-current picture of overlap, so no one has to trust a manual tally.

Target groups include small groups, church and community groups, study groups, volunteer teams, clubs, and small remote teams. This is not an enterprise scheduling platform or a replacement for a personal calendar.

2.1 Reference implementation: When2meet, mapped

Build to this table. Each row is a behaviour users already know from When2meet, restated as a requirement for this product.

When2meet behaviour	Requirement here
Create form: event name, click / drag to pick dates, “No earlier than” and “No later than” dropdowns, time zone dropdown, Create Event button	Copy this form and its labels. No extra required fields.
Two date modes: *Specific Dates* or *Days of the Week*	Both modes are in v1. Days of the Week is for finding a recurring weekly slot.
No account, no dashboard, no event list. The URL is the event.	Same. Losing the link means losing access — say so on the create confirmation.
Per-event sign-in: name + optional password, “only for this event”	Same. It is not an account and must not be presented as one.
One event page, two grids side by side: *Your Availability* (left) and *Group’s Availability* (right)	Same single page. No separate respond page and results page.
Click and drag to paint; drag again to clear. Available = green, unavailable = pink / red	Same gesture, mouse and touch. Painting toggles cells.
Grid cells are 15-minute increments	Same. Hour labels on the axis; four cells per hour.
No Save button — “Click and drag to toggle; saves immediately”	Same. Every paint action saves at once and the page says so.
Group grid is a heatmap: white for 0/N available to dark green for N/N, with that legend shown	Same shading and same legend, e.g. “0/6 Available … 6/6 Available”.
Hover a group cell to see who can and cannot make it, by name	Same. On touch devices, tap does the same job.
No finalize button, no notifications, no calendar invite	Same in v1. The organizer reads the darkest block and announces the time in chat themselves. See Section 8.

> **Intentional improvements over When2meet (only these two in v1):** (1) painting must work well on phone touchscreens, where When2meet is clunky; (2) time zone conversion, date shift, and daylight-saving handling must be provably correct (Section 5), with the viewer’s zone named on screen at all times.

3. How it works — the When2meet flow

1. **Organizer creates an event.** They name it (for example, “Men’s Small Group Online”), click or drag on a calendar to pick the Specific Dates — or switch to Days of the Week — set No earlier than and No later than, confirm the time zone, and press Create Event.
2. **The event page is the share link.** Creating the event lands directly on the event page. The organizer copies that URL and pastes it into the group chat. There is no separate share screen and no admin link.
3. **Each person signs in for this event only.** At the top of the event page they enter their name and, if they wish, a password. The password protects only their answers on this event, so they can edit them later; it is not an account.
4. **They paint their availability.** In Your Availability on the left, they click and drag (or touch and drag) over the 15-minute cells when they are free. Cells turn green; dragging over green cells clears them back to unavailable. There is no Save button — it saves immediately, and the page says so.
5. **Everyone reads the group grid.** Group’s Availability on the right shades each cell by how many people are free. Hovering or tapping a cell lists who is available and who is not. The darkest cells are the meeting times that work for everyone.
6. **The organizer announces the time.** Like When2meet, the product stops there in v1. The organizer picks a darkest block and posts that time back in the group chat.

> **What the participant should feel:** open link, type name, paint, done — under a minute, with the group result already visible while they paint. Any step beyond that is scope creep against the reference implementation.

4. Functional requirements

Priority key: **MUST** = required for the first release. **SHOULD** = include if time allows. **LATER** = not in the first release. **NOT IN V1** = deliberately excluded to stay like When2meet.

4.1 Create an event

- **MUST** Fields, in this order: Event name; date selector; No earlier than (dropdown); No later than (dropdown); Time zone (dropdown, pre-filled from the organizer’s browser); Create Event button.
- **MUST** Date selector has two modes: **Specific Dates** (click / drag on a calendar to select individual dates, at least 14 in one event) and **Days of the Week** (select weekdays, for a recurring slot).
- **MUST** Support a full-day window in one event: No earlier than 12:00 AM through No later than 11:00 PM.
- **MUST** Creating the event generates one unguessable URL and navigates straight to the event page. No account, no email, no dashboard.
- **NOT IN V1** Meeting duration, description form fields, admin links, event editing after creation, and organizer accounts. When2meet has none of these; do not add them silently.

4.2 The event page — one page, two grids

- **MUST** Layout: event name and time zone selector at the top; sign-in (name + optional password) above the grids; **Your Availability** on the left and **Group's Availability** on the right. On a phone the two grids stack, group grid second, with a clear jump link between them.
- **MUST** Both grids share the same axes: dates (or weekdays) across the top, times down the side, labelled hourly, with four 15-minute cells per hour.
- **MUST** The viewer's time zone is named beside the selector at all times, e.g. "Your Time Zone: Pacific Time (Los Angeles)". Changing it re-renders both grids in the new zone.
- **MUST** Before sign-in, Your Availability is shown signed-out / disabled and Group's Availability is already fully visible. Seeing the result must never require signing in.

4.3 Sign in and paint availability

- **MUST** Sign-in is a display name plus an *optional* password, scoped to this event only. Label it exactly in that spirit: name / password are only for this event.
- **MUST** Painting: press and drag paints a run of cells available (green); starting a drag on a green cell clears the run (back to pink / unavailable). Works with mouse and with touch, without the page scrolling while painting on touch.
- **MUST** Cells are 15 minutes. A participant who can make only 45 minutes of an hour must be able to show exactly that.
- **MUST** **No Save button.** Each paint action saves immediately; show the When2meet wording or equivalent: "Click and drag to toggle; saves immediately."
- **MUST** Returning: signing in again with the same name and password loads that person's existing painting for editing. A different spelling creates a different person — warn about this next to the sign-in, because it is When2meet's most common user error.
- **MUST** If a name is password-protected, the wrong password must not load or overwrite that person's answers. If a name has no password, state plainly that anyone with the link can edit it.
- **NOT IN V1** A third "if needed" state, per-participant comments, and checkbox / list entry. When2meet availability is binary — available or not — and painting is the only entry method.

4.4 Group's Availability — the heatmap

- **MUST** Each cell is shaded on a single green scale: white for 0/N available through to the darkest green for N/N available, where N is the number of people who have signed in and painted.
- **MUST** Show the legend beside the grid in When2meet form, e.g. "0/6 Available" (white) to "6/6 Available" (darkest).
- **MUST** Hover (desktop) or tap (touch) on a group cell shows: the date and time in the viewer's zone, the count (e.g. 5/6), and two name lists — Available and Unavailable.
- **MUST** The group grid updates when anyone paints, without recreating the event and, for viewers already on the page, without a manual refresh where practicable.
- **SHOULD** Clicking a name in the participant list highlights that person's availability on the group grid, as a reading aid only.
- **NOT IN V1** A separate ranked "best times" list, score tables, or a finalize / lock-in button. In When2meet the heatmap itself is the ranking — darkest wins — and that directness is the point.

5. Time zone handling — keep When2meet's model, make it correct

When2meet's model is simple: the organizer picks a time zone when creating the event, and each viewer can pick their own time zone at the top of the event page; the grids are then shown in that viewer's zone. Keep that model. What must improve is the correctness underneath it, because a scheduler that looks authoritative while converting wrongly is worse than a form.

- **MUST** **Store everything in UTC.** Every painted cell is stored as a UTC start and end time. Local times are only a display layer.
- **MUST** **Display in the viewer's zone.** Selecting a zone re-renders both grids in that zone, including moving cells to a different calendar day where the conversion requires it. Example: Monday 7:00 PM Pacific appears on Tuesday morning for a WIB viewer, under Tuesday.
- **MUST** **Handle daylight saving time.** Use a maintained time zone database (IANA tz database) via the platform's standard library. Never hard-code offsets such as "Pacific = UTC-8", because the offset changes across a daylight-saving boundary inside a single 14-day event.
- **MUST** **Be explicit when hours shift.** If an event created 9 AM–5 PM in one zone renders as 8 AM–4 PM in the viewer's zone, that is correct behaviour, not a bug — the on-screen zone label is what prevents the surprise, so it must never be hidden or abbreviated to just an offset.
- **MUST** In **Days of the Week** mode, state clearly which zone the weekday columns are defined in, since a Monday evening in one zone can be Tuesday in another.
- **SHOULD** In the hover / tap detail for a group cell, show that cell's time in both the viewer's zone and the organizer's creation zone.

> **Developer note:** Never store a wall-clock time without its zone, and never compare two wall-clock times from different zones directly. Convert to UTC first, compare, then convert back for display. Most bugs in this product category come from breaking that one rule.

Sharing and access (continued from Section 4)

- **MUST** Anyone with the link can view the event and the group grid. Signing in with a name is required only to paint.
- **MUST** No participant account, app install, or email verification at any point.
- **MUST** Works in mobile browsers: most participants will open the link from a phone chat app, and painting (Section 4.3) is the interaction that must survive the small screen.
- **LATER** Optional organizer accounts to list past events. When2meet deliberately has none; add only if users ask for it.

6. Screens and user flows — two screens only

Screen	Purpose and key elements
1. Create Event	Event name; Specific Dates / Days of the Week selector (calendar click-and-drag for specific dates); No earlier than and No later than dropdowns; time zone dropdown; Create Event button. Nothing else is required.
2. Event page	Event name; time zone selector; per-event sign-in (name + optional password); Your Availability grid (left); Group's Availability heatmap with legend (right); hover / tap cell detail with Available and Unavailable name lists. This one page is viewing, responding, and results.

> **Do not split Screen 2.** There are no separate Share, Respond, Results, or Finalized screens. That is not When2meet, and it is not this product. The single event page is the design.

Organizer flow

Create Event → land on the event page → sign in with their name and paint their own availability → copy the page URL → paste it into the group chat → revisit the same link to watch the heatmap darken → announce a darkest time in chat.

Participant flow

Tap link in chat → see the group grid straight away → check the time zone shown is theirs (change it if not) → type name (and optional password) → drag-paint free times — saved immediately → close the page → return later, sign in with the same name and password, and repaint to edit.

7. Data model and technical notes

Entity	Key fields
Event	id / URL token, name, mode (specific dates / days of week), selected dates or weekdays, no-earlier-than, no-later-than, creation time zone, created at
Participant (per event)	id, event id, display name, optional password hash (for this event only), created at, updated at
Availability	participant id, cell start (UTC), cell end (UTC) — 15-minute cells generated from the event's defined grid, state is available or not available (absence of a cell = not available)

- Availability cells come only from the event's defined grid, so every participant answers against the identical cell set in UTC, whatever zone they painted in.
- Painting saves per drag action, not on a submit. Design the write path for many small saves, and make a repeated save of the same cells harmless.
- Editing is re-sign-in, not a separate edit screen: the same name + password loads the same participant record. Never create a second participant for an exact name match.
- Non-functional requirements: the page must load usefully on a typical mobile connection; a 14-day grid will need horizontal scrolling on phones, with date headers and the time axis staying visible; shading must not be the only signal — the hover / tap name lists and the x/N count are the text alternative; all labels are real, selectable text.
- Privacy, stated plainly on the event page: names and availability are visible to anyone with the link. That visibility is the product's purpose. Collect no email addresses in v1.

8. Scope: first release, later, and non-goals

MVP — a faithful When2meet clone

- Create Event: name, Specific Dates or Days of the Week, No earlier than / No later than, time zone
- One shareable URL; no accounts, no dashboard
- Single event page: Your Availability + Group's Availability
- Name + optional per-event password sign-in
- Drag-paint in 15-minute cells, saves immediately, no Save button
- Green heatmap with 0/N to N/N legend; hover / tap shows names
- UTC storage, per-viewer zone display, date-shift and DST handling
- Touch painting that works on phones

Later — only if users ask

- Finalize a time in-product, with a copyable multi-zone announcement
- Ranked best-times list alongside (never instead of) the heatmap
- Calendar file (.ics) download for the chosen time
- "If needed" as a third availability state
- Organizer accounts, event editing, and event history
- Reminders to people who have not yet painted

Deliberately out of v1 — because When2meet leaves them out

- **No finalize workflow.** When2meet ends at the heatmap; the organizer announces the time elsewhere. Adding a finalize step changes the product's character, so it is a Later decision, not a missing MVP feature.
- **No accounts or event management.** No sign-up, no dashboard, no event list, no admin link.
- **No notifications, chat, video calling, payments, or attendance tracking.** The meeting and its announcement happen elsewhere.
- **No poll-style shortlist mode.** Proposing a few fixed times for yes / no voting is a different product (closer to Rallly or Doodle). This product is the open painted grid.
- **No calendar integration in v1.** Participants paint from memory, as they do in When2meet.

9. Acceptance criteria and open questions

Acceptance criteria — the build is done when all of these pass

1. An organizer can create an event for 14 specific dates (Nov 2–15) covering the full day, using the labels No earlier than and No later than, and lands directly on an event page with a single shareable URL.
2. An organizer can instead create a Days of the Week event for a recurring slot.
3. A participant can sign in with only a name (password left blank), drag-paint availability in 15-minute cells, and leave the page with no Save action — and their painting is already stored.
4. Painting works with touch on a phone without the page scrolling, including painting only 45 minutes of an hour.
5. Signing in again with the same name and password loads the same person's painting for editing; a protected name cannot be loaded with the wrong password.
6. Group's Availability shows the 0/N to N/N green legend, and hovering or tapping any cell shows the correct count and the Available / Unavailable name lists.
7. A participant in WIB and a participant in Pacific Time, opening the same event, each see both grids in their own zone with the zone named on screen; a Monday 7:00–8:00 PM Pacific painting appears on the correct WIB day and hour.
8. An event spanning a daylight-saving change shows correct local hours on both sides of the change.
9. The event page contains no account sign-up, no Save button, no ranked list, and no finalize button — their absence is a pass condition, not an omission.

Open questions for the product owner

- Password behaviour: copy When2meet exactly (optional, per event, reusable by the same person on later events is out of scope) — or require a password? Recommendation: copy When2meet.
- How long should events and their data be kept, given there are no accounts to delete them with?
- Week start in the date selector: Sunday or Monday first, or follow the viewer's locale? When2meet lets this vary; pick one rule and state it on the create form.
- If a Later phase is commissioned, which single item from the Later list comes first — finalize, or .ics download?

> **Success measure:** A real small group, split across at least two time zones, schedules a real meeting using only the link — painting, reading the darkest block, and announcing that time — with no side conversation needed to translate times, and no member reporting that the announced time was wrong for them.